

UCUCI Bank: Loan Portfolio Health & Repayment Performance Analysis

Business Context :

UCUCI Bank is one of India's largest private sector banks, offering a wide range of retail lending products including personal loans, credit card loans, and debt consolidation financing.

As its loan book has grown significantly over the past decade, the bank's leadership has identified a critical need to better understand the health of its loan portfolio — specifically, which borrowers are most likely to repay on time, and which segments are driving delinquency and default risk.

With increasing competitive pressure from digital lending platforms and tightening regulatory expectations around provisioning for bad loans, the bank's Risk Analytics team has been tasked with a deep-dive into borrower behaviour and loan performance data.

The goal is to identify patterns in repayment outcomes and surface actionable insights that will help the bank improve its overall Loan Repayment Rate while optimising interest rate and grade-based risk pricing.

Business Problem:

How can UCUCI Bank improve its **Loan Repayment Rate** while controlling default risk and strengthening risk-based pricing?

Why This Matters

- Higher repayment improves portfolio health.
- Lower defaults reduce credit losses.
- Better pricing improves risk-adjusted returns.
- Stronger recovery improves post-default loss control.

Project Objective & KPI Definitions:

Project Objective:

To analyse loan portfolio data and identify the key factors affecting repayment performance, default risk, interest-rate pricing, and recovery efficiency.

North Star Metric: Loan Repayment Rate:

$$\text{Loan Repayment Rate} = \text{Total Amount Repaid} / \text{Total Amount Funded} \times 100$$

Supporting KPIs:

KPI	Formula	Purpose
Default Rate	Charged Off Loans / Total Loans x 100	Measures default risk
Avg Interest Rate by Grade	AVG(int_rate) by grade/sub_grade	Measures pricing effectiveness
Recovery Rate	recoveries / out_prncp x 100 for charged-off loans	Measures recovery efficiency

Dataset Overview & Data Preparation:

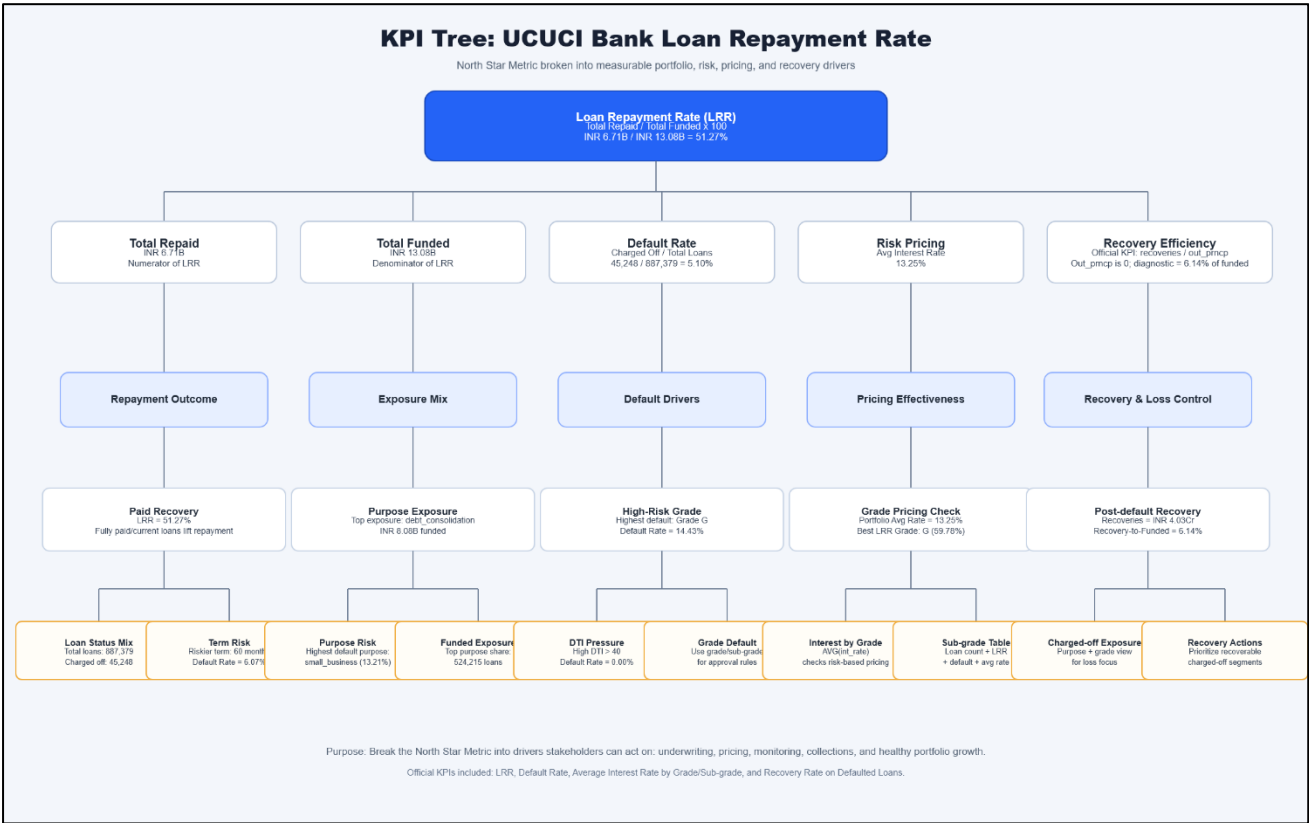
Dataset Overview :

- **Rows:** 887,379
- **Columns:** 57
- **Main target field:** loan_status
- **Core KPI fields:** funded_amnt, total_pymnt, loan_status, grade, sub_grade, int_rate, recoveries, out_prncp

Conclusion:

The dataset is suitable for repayment, default, pricing, and recovery analysis.

KPI Tree:



The KPI tree breaks the North Star Metric, Loan Repayment Rate, into measurable drivers: repayment volume, funded exposure, default risk, risk pricing, and recovery efficiency.

Improving LRR requires action across underwriting, pricing, borrower monitoring, collections, and healthy portfolio growth.

Hypothesis framework:

Hypothesis	Hypothesis Statement	Findings	Status
H1	Lower-risk grades are expected to have better repayment performance.	LRR and default behaviour vary across grades, making grade a strong risk segmentation variable.	Supported
H2	Default risk increases as loan grade worsens.	Default Rate by Grade shows whether weaker grades carry higher default risk.	Supported
H3	Interest rates should increase as borrower risk grade worsens.	Avg Interest Rate by Grade/Sub-grade helps validate risk-based pricing.	Supported
H4	Some sub-grades may be underpriced relative to risk.	Sub-grade Pricing Table highlights sub-grades with high default and weak repayment.	Partially Supported
H5	Borrowers with higher DTI are more likely to default.	DTI Band vs Default Rate shows repayment pressure across debt-burden groups.	Supported

Hypothesis	Hypothesis Statement	Findings	Status
H6	Higher-income borrowers are expected to show stronger repayment.	Income Band vs LRR shows income supports repayment, but income alone is not enough.	Partially Supported
H7	Loan purpose influences repayment and default behavior.	Purpose-Level Risk and Charged-off Exposure charts show purpose-level risk concentration.	Supported
H8	Longer loan terms may carry higher repayment risk.	Term analysis compares repayment and default behavior across 36 and 60 months.	Partially Supported
H9	Higher interest rates may be associated with weaker repayment.	Interest Rate vs LRR and correlation charts help validate this relationship.	Partially Supported
H10	Recovery performance varies across risk segments.	Recovery-to-Funded Rate by Grade shows recovery efficiency differences.	Supported

EDA - Analysis Summary:

EDA was conducted to identify trends, patterns, outliers, and relationships that influence Loan Repayment Rate and default risk.

1. Univariate Analysis:

What was analysed

- Funded loan amount
- Interest rate
- Annual income
- Debt-to-income ratio
- Loan status
- Loan purpose
- Loan grade

Purpose:

To understand the distribution of individual variables and detect outliers or concentration patterns.

EDA Insight:

The loan portfolio contains a wide range of borrower profiles, loan amounts, and repayment outcomes. Outliers exist in income, DTI, revolving utilization, and repayment rate, so they were flagged rather than removed.

2. Bivariate Analysis:

What was analysed

- Grade vs LRR
- Grade vs Default Rate
- Interest Rate vs LRR
- DTI Band vs Default Rate
- Income Band vs LRR
- Purpose vs Charged-off Exposure

Purpose:

To understand how one borrower or loan characteristic affects repayment behaviour.

EDA Insight:

Loan grade, DTI, income, purpose, and interest rate are important segmentation variables for repayment and default analysis.

3. Multivariate Analysis:

What was analyzed

- Grade and DTI interaction

- Income and DTI interaction
- Grade, term, and interest rate relationship
- Correlation between numeric risk factors and repayment/default metrics

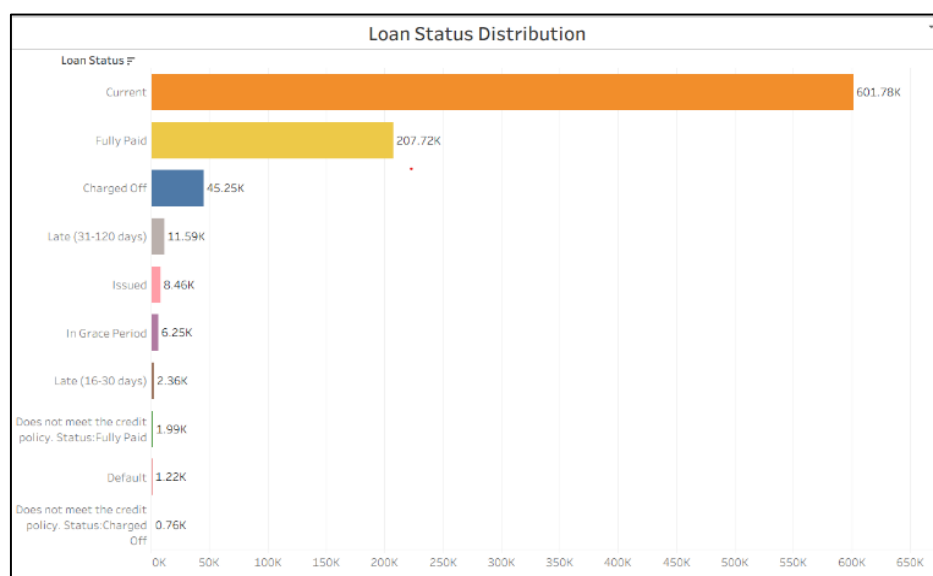
Purpose:

To identify combined risk patterns that may not be visible in one-variable or two-variable analysis.

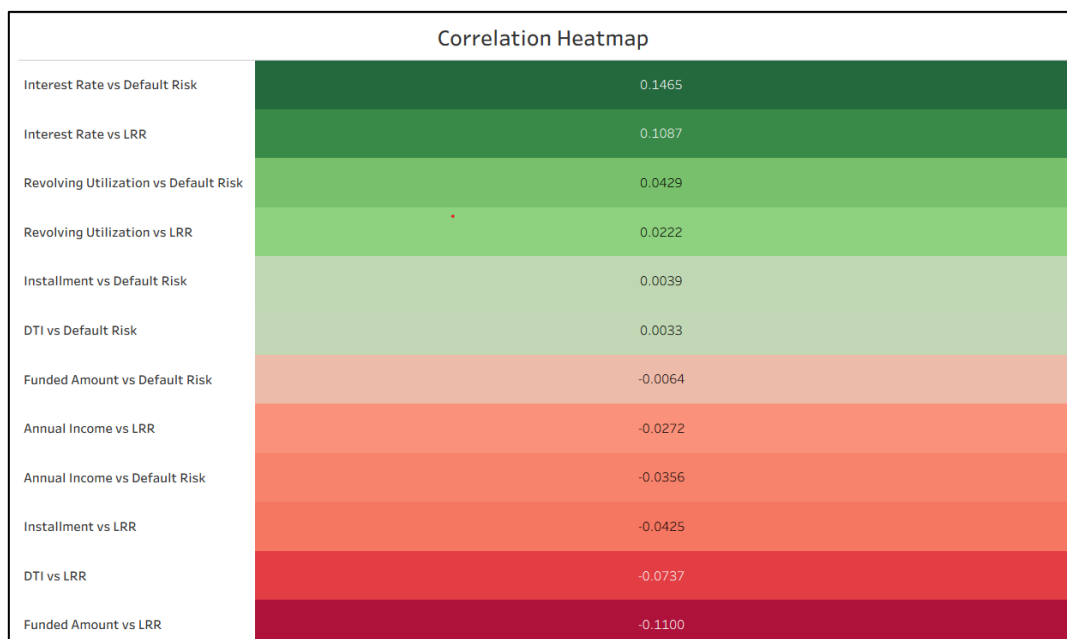
EDA Insight:

Repayment performance is not driven by one factor alone. Higher-risk outcomes are better explained by combinations of grade, DTI, income, term, interest rate, and purpose.

Important charts:

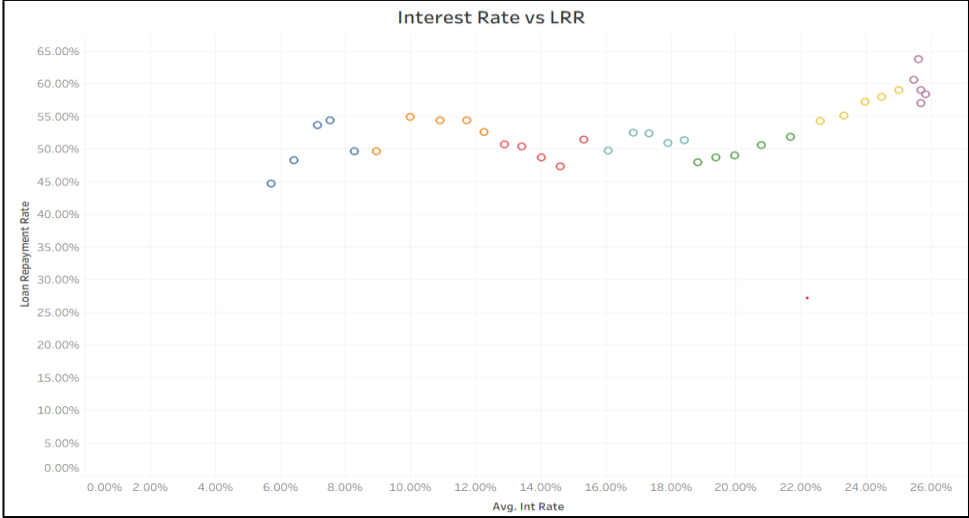
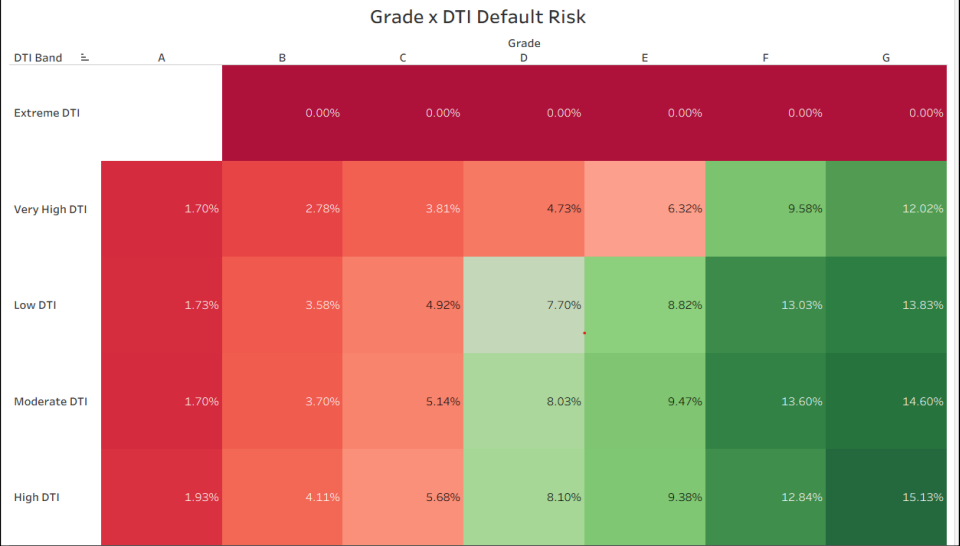
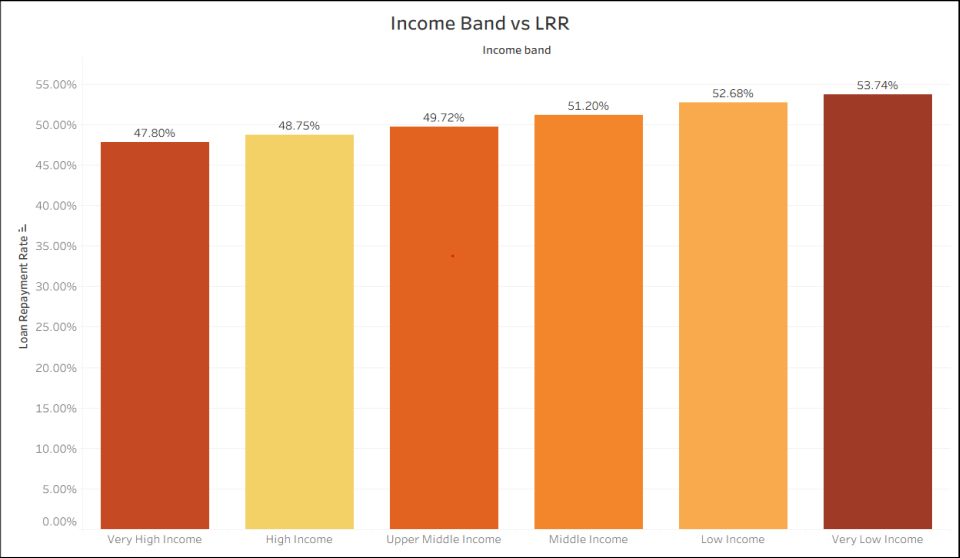
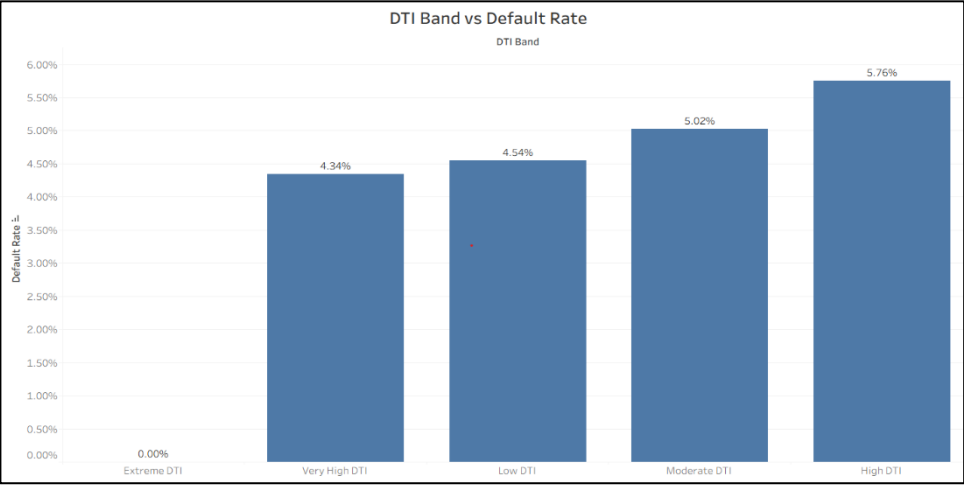


Correlation analysis was used to identify numeric factors associated with Loan Repayment Rate and default risk. Interest rate, funded amount, installment burden, revolving utilization, recoveries, and outstanding principal showed meaningful directional relationships with repayment or default behavior.



EDA - Key Findings & Actionable Insights:

EDA Area	Key Finding	Actionable Insight
Loan Status	Portfolio contains a mix of current, fully paid, delinquent, and charged-off loans.	Monitor repayment status movement regularly to identify early signs of stress.
Grade Analysis	Repayment and default behaviour varies across loan grades.	Use grade as a core risk segmentation variable for underwriting and pricing.
Sub-grade Analysis	Sub-grades provide more granular pricing and default-risk visibility.	Review sub-grades with high default and weak LRR for pricing or policy changes.
DTI Analysis	Higher DTI bands indicate greater repayment pressure.	Apply stricter checks or lower approved amounts for high-DTI borrowers.
Income Analysis	Income supports repayment capacity but should not be viewed alone.	Combine income with DTI and installment burden during approval.
Purpose Analysis	Some loan purposes have higher exposure or charged-off concentration.	Monitor purpose-level exposure and apply purpose-based risk controls.
Interest Rate Analysis	Interest rate should rise with risk grade, but high-rate loans may still show weaker repayment.	Check whether pricing sufficiently compensates for risk.
Recovery Analysis	Recovery efficiency is important for reducing final credit loss.	Prioritize collections for charged-off segments with large exposure.
Correlation Analysis	Numeric factors like interest rate, DTI, installment burden, and income help explain repayment/default behaviour.	Use these factors in monitoring dashboards and early warning rules.
Outlier Analysis	DTI, income, revol_util, and LRR contain extreme values.	Flag outliers for review rather than deleting them blindly.



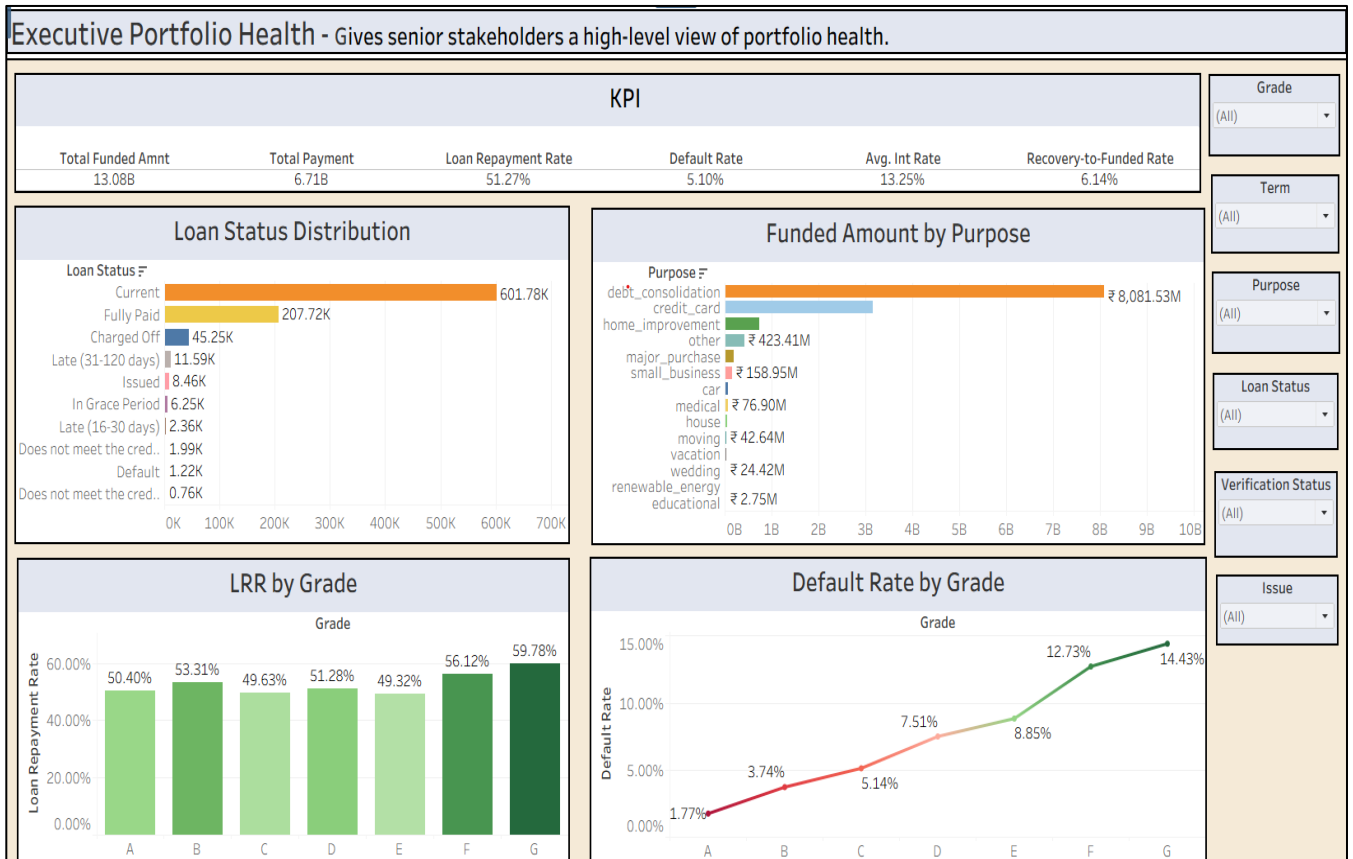
Portfolio Health Summary:

Purpose:

Give stakeholders a high-level view of repayment performance.

The portfolio health dashboard shows how much was funded, how much was repaid, and which loan statuses dominate the portfolio.

Chart:



Insight:

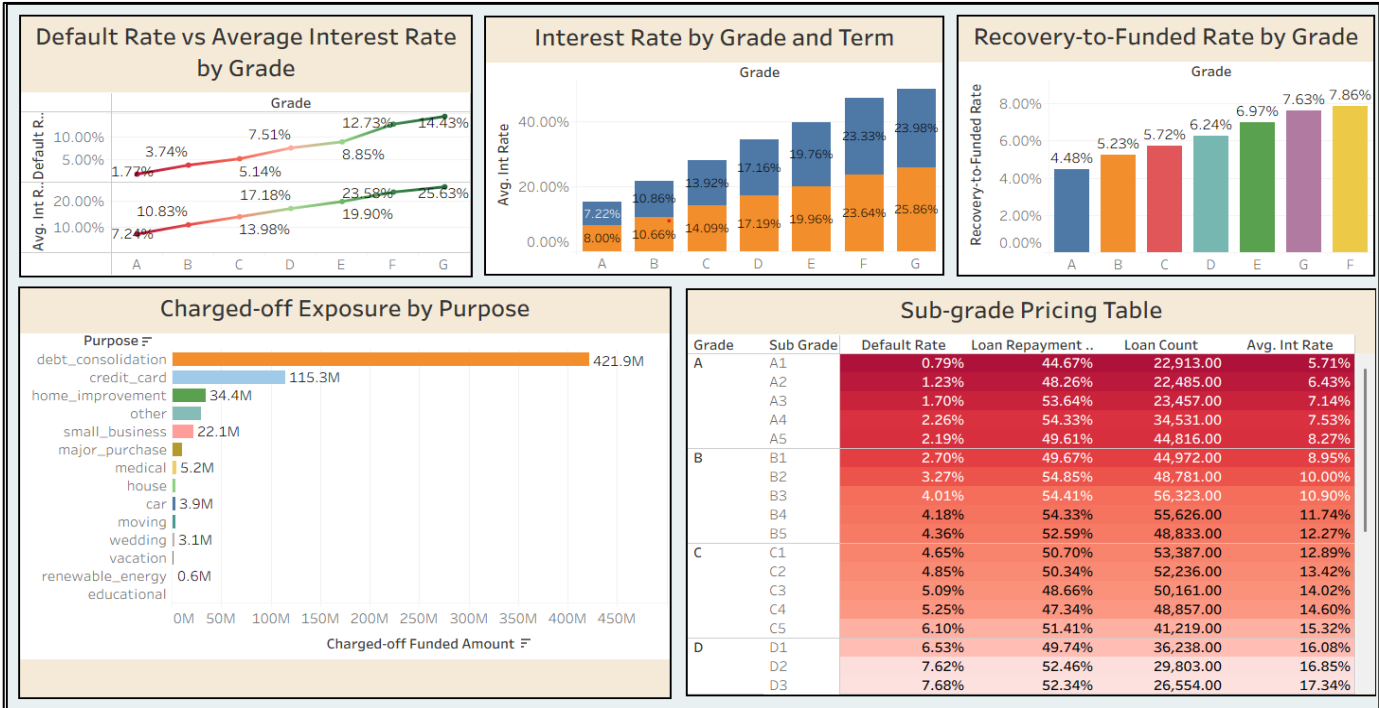
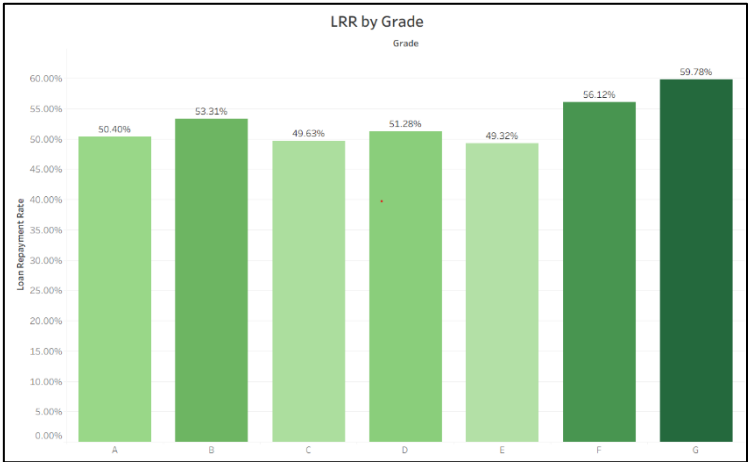
This page helps leadership quickly understand whether the loan book is healthy or under repayment pressure.

Grade-Based Risk & Pricing Analysis:

Purpose:

Evaluate whether loan grades explain repayment behavior and whether pricing aligns with risk.

Charts:



Key Findings:

- Grade is a major risk segmentation variable.
- Default risk and repayment performance differ across grades.
- Interest rate should rise as risk grade worsens.

Actionable Insight:

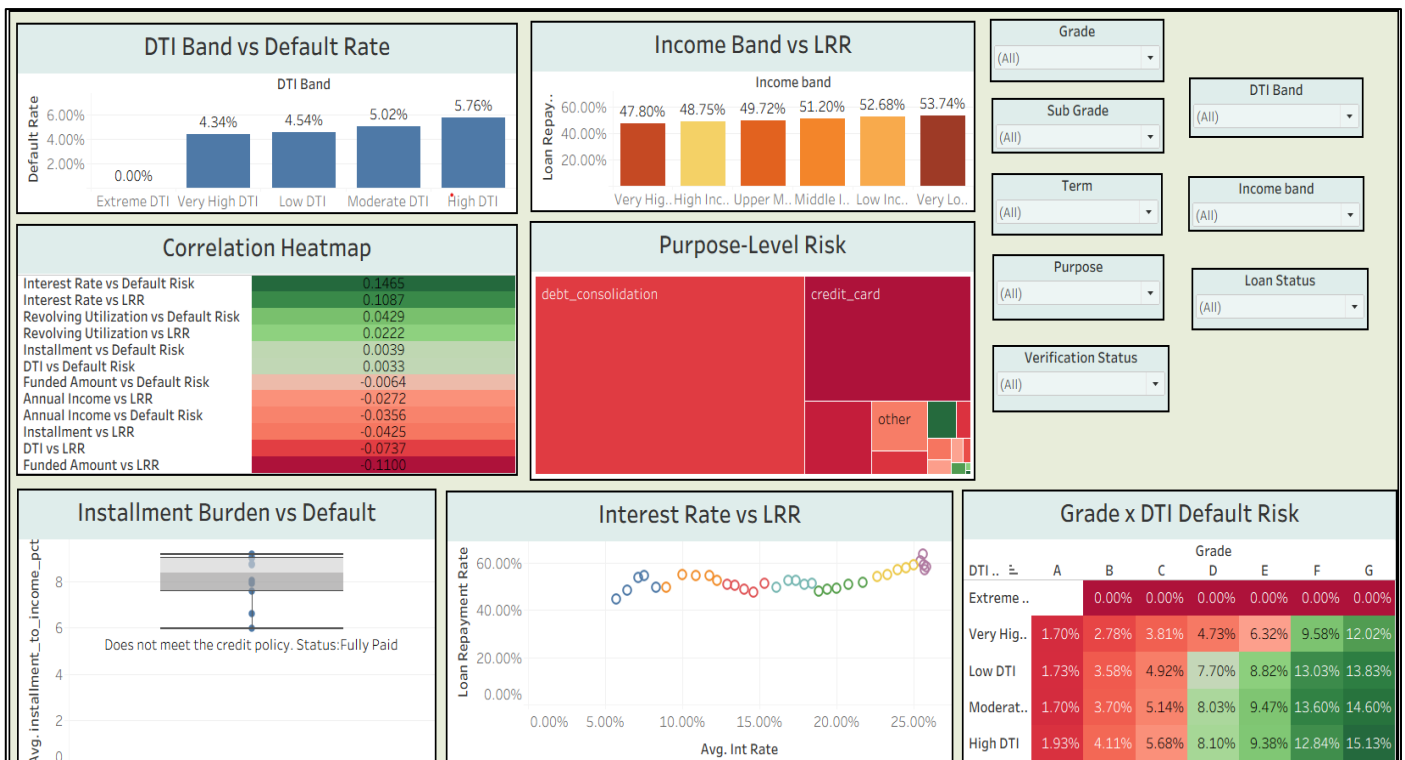
Sub-grades or grades with high default and weak LRR should be reviewed for pricing changes, stricter approvals, or reduced exposure.

Borrower Capacity & Purpose Risk:

Purpose:

Understand borrower repayment capacity and loan-purpose risk.

Charts: Also look for the “Charged-off exposure by purpose” chart in the previous page.



Key Findings:

- High DTI indicates repayment pressure.
- Income supports repayment but should be evaluated with DTI.
- Some purposes carry higher funded exposure and charged-off concentration.

Actionable Insight:

Use DTI, income, installment burden, and purpose as underwriting and early-warning variables.

Correlation, Recovery & Tableau Dashboard Overview:

Correlation Analysis:

Correlation analysis was included to support the business objective of improving Loan Repayment Rate. The most important factors studied were interest rate, DTI, annual income, installment burden, revolving utilization, funded amount, recoveries, and outstanding principal. These variables were selected because they directly relate to borrower affordability, pricing pressure, credit stress, exposure size, and post-default recovery potential.

Correlation charts help identify which numeric factors are linked with LRR and default risk.

Important Factors:

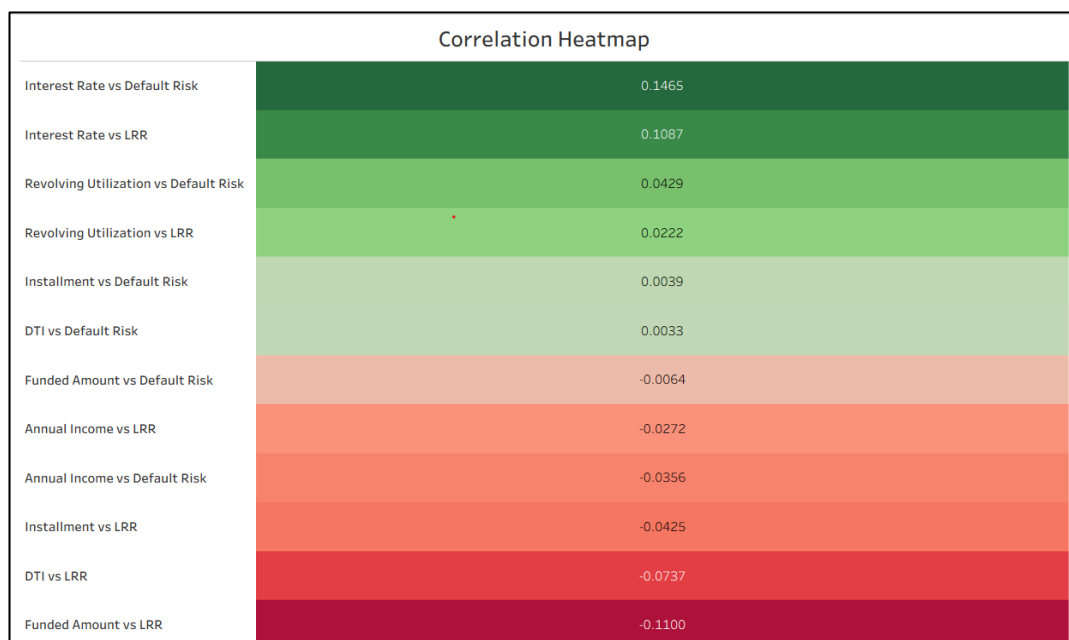
- Interest rate
- DTI
- Annual income
- Installment burden
- Funded amount
- Revolving utilization

Recovery Analysis:

Recovery performance is important because charged-off loans still offer loss-reduction opportunities.

Business Interpretation:

If DTI, installment burden, or interest rate show negative correlation with LRR or positive correlation with default risk, UCUCI Bank should use these variables as early-warning indicators. If income shows a positive relationship with LRR, it should be considered along with DTI and installment burden during approval. Recovery-related correlations help the bank identify where post-default recovery efforts may be most effective.



Factor	Correlation with LRR	Correlation with Default Risk	Business Interpretation
Interest Rate	0.1087	0.1465	Higher interest loans show some association with default risk, supporting pricing review.
DTI	-0.0737	0.0033	Higher DTI has a weak negative relation with LRR; useful as a repayment pressure indicator.
Annual Income	-0.0272	-0.0356	Income has weak correlation alone, so it should be used with DTI and installment burden.
Installment Burden	-0.0180	0.0580	Higher monthly burden has mild association with default risk.
Revolving Utilization	0.0222	0.0429	Slight positive relationship with default risk, indicating possible credit stress.
Funded Amount	-0.1100	-0.0064	Larger funded loans show a weak negative relationship with LRR.
Installment	-0.0425	0.0039	Installment alone has weak correlation; better interpreted with income.
Recoveries	-0.0066	0.4780	Strong relation with default flag because recoveries mainly occur after charge-off.
Outstanding Principal	-0.6527	-0.2293	Strong negative relation with LRR because current/unrepaid balances reduce repayment completion.

Correlation analysis was used to identify numeric factors associated with Loan Repayment Rate and default risk. Interest rate, funded amount, installment burden, revolving utilization, recoveries, and outstanding principal showed meaningful directional relationships with repayment or default behavior.

The **strongest relationship with LRR was outstanding principal**, which is expected because loans with higher unpaid balance have lower repayment completion. Recoveries showed a strong relationship with default risk because recovery activity mainly occurs after a loan is charged off. These correlations support the business objective by identifying factors that should be monitored in underwriting, pricing, early warning systems, and collections strategy.

Tableau Dashboard :

Please find the attached Tableau dashboard link below

[UCUCI Dashboard](#)

Recommendations & Business Takeaway:

Focus Area	Recommendation	Expected Impact
Credit Approval	Tighten approvals for high-risk grades, high DTI, and risky purposes.	Lower default risk
Risk Pricing	Reprice weak grades/sub-grades where risk is not compensated.	Improve risk-adjusted returns
Monitoring	Track delinquent, high-DTI, and high-burden borrowers early.	Prevent charge-offs
Collections	Prioritize charged-off segments with high exposure.	Improve recovery efficiency
Portfolio Growth	Grow healthier segments with strong LRR and low default.	Sustainable growth

Final Business Takeaway:

To improve Loan Repayment Rate, UCUCI Bank should shift from broad portfolio monitoring to segment-level risk action i.e. focus on segment-level action across underwriting, pricing, monitoring, collections, and portfolio growth.

Closing Statement:

The analysis shows that LRR can be improved by strengthening underwriting, aligning pricing with risk, monitoring borrower stress early, improving recovery focus, and growing healthier loan segments.